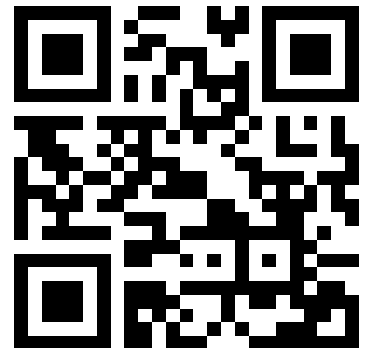


# Advanced Micro-Controller Systems

- Introduction, Microcontroller basics
- 32-bit Microcontroller technology
- • ARM Cortex M Architecture: CM0 CM3 CM4 CM7
- Cortex M Microcontroller Instruction Set
- Exception Handling, Interrupts
- Real-time Operating Systems (RTOS)
- Memory Management, protected memory, MPU
- Implementing Calculations, Data Formats, FPU
- I/O, Device drivers
- Application Design
- Case Studies, Advanced Design Patterns

***<https://skript.eit.h-da.de/ams/>***



# ARM Family of Microcontrollers

## Cortex M Product Line

### **Cortex M0**

|                              |       |
|------------------------------|-------|
| Instruction Set Architecture | (ISA) |
| Registers                    |       |
| Memory Model                 |       |

### **Cortex M3**

|                              |           |
|------------------------------|-----------|
| Instruction Set Architecture | (CM3 ISA) |
| I/O Subsystem, Programming   | (CMSIS)   |
| Interrupt Controller         | (NVIC)    |
| Operation Modes              |           |

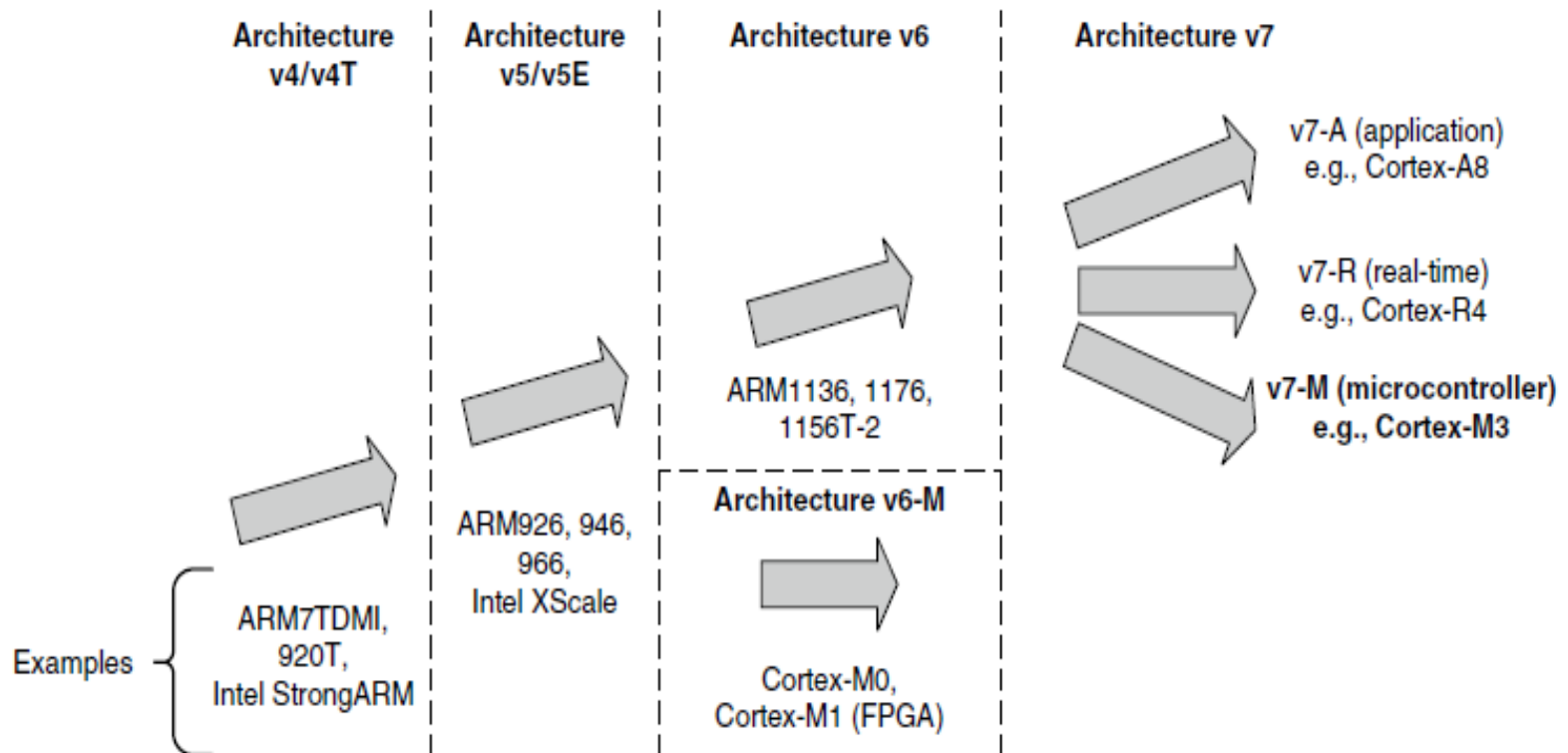
### **Cortex M4**

|                                 |       |
|---------------------------------|-------|
| Memory Protection Unit          | (MPU) |
| Digital signal Processing       | (DSP) |
| Floating Point computation Unit | (FPU) |

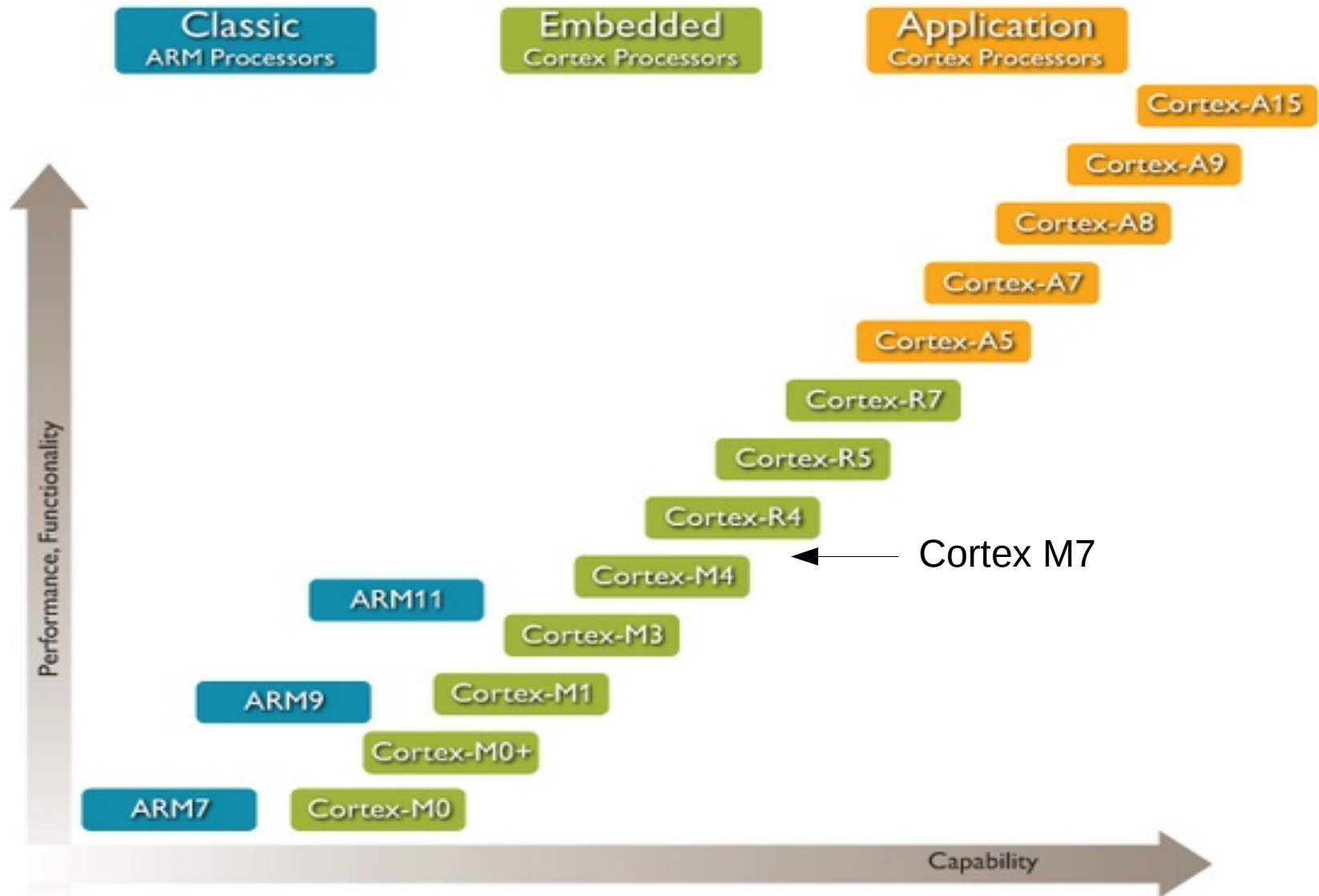
### **Cortex M7**

|                      |                           |
|----------------------|---------------------------|
| Adds memory caching  |                           |
| Improved performance | (Up to a factor of 3 !!!) |

# ARM Architecture Development



# ARM Microcontrollers



## ARM Cortex-M Optional Components <sup>[6][7]</sup>

| ARM Cortex-M              | SysTick Timer | Bit Banding             | Memory Protection Unit (MPU) |
|---------------------------|---------------|-------------------------|------------------------------|
| Cortex-M0 <sup>[1]</sup>  | Optional      | Optional <sup>[9]</sup> | No                           |
| Cortex-M0+ <sup>[2]</sup> | Optional      | Optional <sup>[9]</sup> | Optional                     |
| Cortex-M1 <sup>[3]</sup>  | Optional      | No                      | No                           |
| Cortex-M3 <sup>[4]</sup>  | Yes           | Optional                | Optional                     |
| Cortex-M4 <sup>[5]</sup>  | Yes           | Optional                | Optional                     |

Cortex M7

Yes

NO !

Yes

# ARM Cortex Instruction Sets

ARM Cortex-M Instruction Sets <sup>[6][7]</sup>

| ARM Cortex-M              | Thumb  | Thumb-2 | Hardware Multiply | Hardware Divide | Saturated Math | DSP Extensions | Floating Point | ARM Architecture        |
|---------------------------|--------|---------|-------------------|-----------------|----------------|----------------|----------------|-------------------------|
| Cortex-M0 <sup>[1]</sup>  | Most   | Subset  | 1 or 32 cycle     | No              | No             | No             | No             | ARMv6-M <sup>[6]</sup>  |
| Cortex-M0+ <sup>[2]</sup> | Most   | Subset  | 1 or 32 cycle     | No              | No             | No             | No             | ARMv6-M <sup>[6]</sup>  |
| Cortex-M1 <sup>[3]</sup>  | Most   | Subset  | 3 or 33 cycle     | No              | No             | No             | No             | ARMv6-M <sup>[6]</sup>  |
| Cortex-M3 <sup>[4]</sup>  | Entire | Entire  | 1 cycle           | Yes             | Yes            | No             | No             | ARMv7-M <sup>[7]</sup>  |
| Cortex-M4 <sup>[5]</sup>  | Entire | Entire  | 1 cycle           | Yes             | Yes            | Yes            | Optional       | ARMv7E-M <sup>[7]</sup> |

Cortex M7    Entire    Entire    1 cycle    Yes    Yes    Yes    32 (64bit)



|       |       |       |        |       |       |      |      |
|-------|-------|-------|--------|-------|-------|------|------|
| VABS  | VADD  | VCMP  | VCMPPE | VCVT  | VCVTR | VDIV | VLDM |
| VLDR  | VMLA  | VMLS  | VMOV   | VMRS  | VMSR  | VMUL | VNEG |
| VNMLA | VMMLS | VNMUL | VPOP   | VPUSH | VSQRT | VSTM | VSTR |
| VSUB  | VFMA  | VFMS  | VFNMA  | VFNMS |       |      |      |

## Cortex-M4 FPU

|        |        |         |         |         |         |        |         |
|--------|--------|---------|---------|---------|---------|--------|---------|
| PKH    | QADD   | QADD16  | QADD8   | QASX    | QDADD   | QDSUB  | QSAX    |
| QSUB   | QSUB16 | QSUB8   | SADD16  | SADD8   | SASX    | SEL    | SHADD16 |
| SHADD8 | SHASX  | SHSAX   | SHSUB16 | SHSUB8  | SMLABB  | SMLABT | SMLATB  |
| SMLATT | SMLAD  | SMLALBB | SMLALBT | SMLALTB | SMLALTT | SMLALD | SMLAWB  |
| SMLAWT | SMLS   | SMLS    | SMMLA   | SMMLS   | SMMUL   | SMUAD  | SMULBB  |

|        |        |        |       |       |       |
|--------|--------|--------|-------|-------|-------|
| ADC    | ADD    | ADR    | AND   | ASR   | B     |
| CLZ    | BFC    | BF     | BIC   | CDP   | CLREX |
| CBNZ   | CBZ    | CMN    | CMP   | DBG   | EOR   |
| LDMIA  | LDMDB  | LDR    | LDRB  | LDRBT | LDRD  |
| LDREX  | LDREXB | LDREXH | LDRH  | LDRHT | LDRSB |
| LDRSBT | LDRSHT | LDRSH  | LDRT  | MCR   | LSL   |
| LSR    | MCRR   | MLS    | MLA   | MOV   | MOVT  |
| MRC    | MRRC   | MUL    | MVN   | NOP   | ORN   |
| ORR    | PLD    | PLDW   | PLI   | POP   | PUSH  |
| RBIT   | REV    | REV16  | REVSH | ROR   | RRX   |

|       |       |       |       |      |
|-------|-------|-------|-------|------|
| BKPT  | BLX   | ADC   | ADD   | ADR  |
| BX    | CPS   | AND   | ASR   | B    |
| DMB   |       | BL    |       | BIC  |
| DSB   | CMN   | CMP   |       | EOR  |
| ISB   | LDR   | LDRB  | LDM   |      |
| MRS   | LDRH  | LDRSB | LDRSH |      |
| MSR   | LSL   | LSR   | MOV   |      |
| NOP   | REV   | MUL   | MVN   | ORR  |
| REV16 | REVSH | POP   | PUSH  | ROR  |
| SEV   | SXTB  | RSB   | SBC   | STM  |
| SXTH  | UXTB  | STR   | STRB  | STRH |
| UXTH  | WFE   | SUB   | SVC   | TST  |
| WFI   | YIELD |       |       |      |

## Cortex-M0/M0+/M1

|       |        |        |
|-------|--------|--------|
| RSB   | SBC    | SBFX   |
| SDIV  | SEV    | SMLAL  |
| SMULL | SSAT   | STC    |
| STMIA | STMDB  | STR    |
| STRB  | STRBT  | STRD   |
| STREX | STREXB | STREXH |
| STRH  | STRHT  | STRT   |
| SUB   | SXTB   | SXTH   |
| TBB   | TBH    | TEQ    |
| TST   | UBFX   | UDIV   |
| UMLAL | UMULL  | USAT   |
| UXTB  | UXTH   | WFE    |
| WFI   | YIELD  | IT     |

## Cortex-M3

|         |         |
|---------|---------|
| SMULBT  | SMULTT  |
| SMULTB  | SMULWT  |
| SMULWB  | SMUSD   |
| SSAT16  | SSAX    |
| SSUB16  | SSUB8   |
| SXTAB   | SXTAB16 |
| SXTAH   | SXTB16  |
| UADD16  | UADD8   |
| UASX    | UHADD16 |
| UHADD8  | UHASX   |
| UHSAX   | UHSUB16 |
| UHSUB8  | UMAAL   |
| UQADD16 | UQADD8  |
| UQASX   | UQSAX   |
| UQSUB16 | UQSUB8  |
| USAD8   | USADA8  |
| USAT16  | USAX    |
| USUB16  | USUB8   |
| UXTAB   | UXTAB16 |
| UXTAH   | UXTB16  |

## Cortex-M4

VABS

VADD

VCMP

VCMP

VCVT

VCVTR

VCVTR

## Floating Point

CVTA

CVTN

PKHBT

PKHTB

QADD

QADD16

QADD8

QASX

QDADD

QDSUB

QSAX

QSUB

QSUB16

QSUB8

QSUB8

QSUB8

QSUB8

SEL

SHADD16

SHADD8

SHASX

SHSAX

SHSAX

SHSAX

SHSAX

SMLABT

SMLATB

SMLATT

SMLAD

SMLADX

SMLADX

SMLADX

SMLALT

SMLALT

SMLALD

SMLALDX

SMLAWB

SMLAWT

SMLAWT

SMLAWT

SMLSLDX

SMLSLDX

## DSP (SIMD, fast MAC)

ADC

ADD

ADR

AND

ASR

B

BFC

BFI

BIC

CBNZ

CBZ

CDP

CDP2

CLREX

CLZ

CMN

CMP

DBG

EOR

LDC

LDC2

LDMIA

LDMDB

LDR

LDRB

LDRBT

LDRD

LDRD

LDRD

LDRH

LDRH

LDRHT

LDRSB

LDRSB

LDRSB

LSL

LSL

LSR

MCR

MCR

MCR

MLA

MLA

MLS

MOV

MOV

MOV

NRR

NRR

MRR2

MUL

MVN

NOP

ORN

ORR

PLD

PLI

POP

PUSH

RBIT

REV

REV16

REVSH

ROR

RRX

RSB

SBC

SBFX

SDIV

SEV

SMLAL

SMULL

SSAT

STC

STC2

STMIA

STMDB

STR

STRB

STRBT

STRD

STREX

STREXB

STREXH

STRH

STRHT

STRT

SUB

SXTB

SXTB

TBB

TBH

TEQ

TST

UBFX

UDF

UDIV

UMLAL

UMULL

USAT

UXTB

UXTH

WFE

WFI

YIELD

IT

Advanced data processing  
bit field manipulations

ADC

ADD

ADR

AND

ASR

B

BIC

BKPT

BL

BLX

BX

CMN

CMP

CPS

DMB

EOR

DSB

ISB

LDMIA

LDR

REV16

REVSH

ROR

RSB

SBC

SEV

STMIA

STR

STRB

STRH

SUB

SVC

SXTB

SXTB

TST

UDF

UXTB

UXTH

WFE

WFI

YIELD

General data processing  
I/O control tasks

Cortex-M0+

Cortex-M3

Cortex-M4  
Cortex-M7Cortex-M4  
FPUCortex-M7  
FPU

VFMA

VFMS

VFNMA

VFNMS

VLDM

VLDR

VMLA

VMLS

VMOV

VMRS

VMSR

VMUL

VNEG

VNMLA

VNMLS

VNMUL

VPOP

VPUSH

VSQRT

VSTM

VSTR

VSUB

VCVTP

VCVTM

VMAXNM

VMINNM

VRINTA

VRINTN

VRINTP

VRINTM

VRINTX

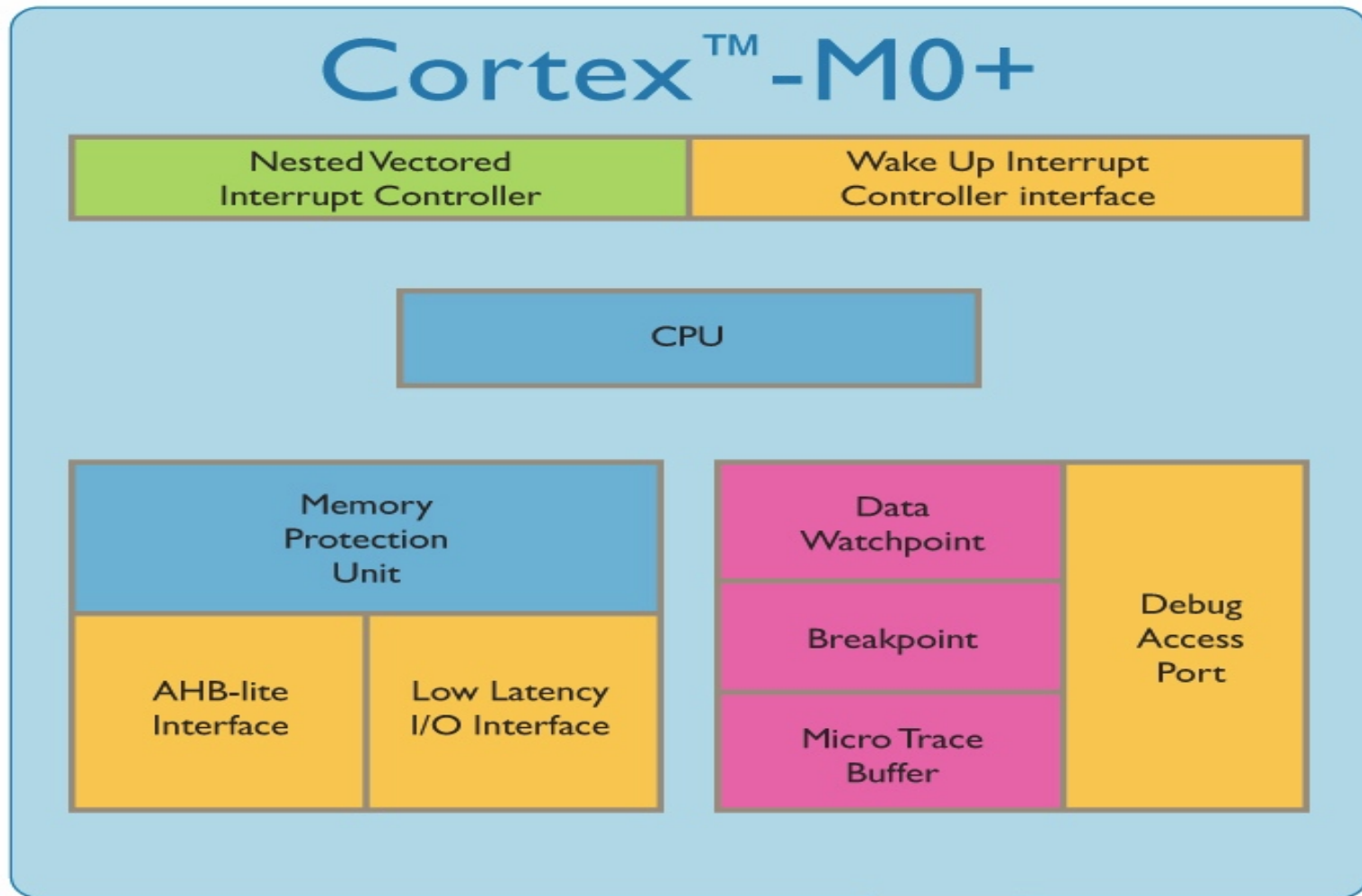
VRINTZ

VRINTR

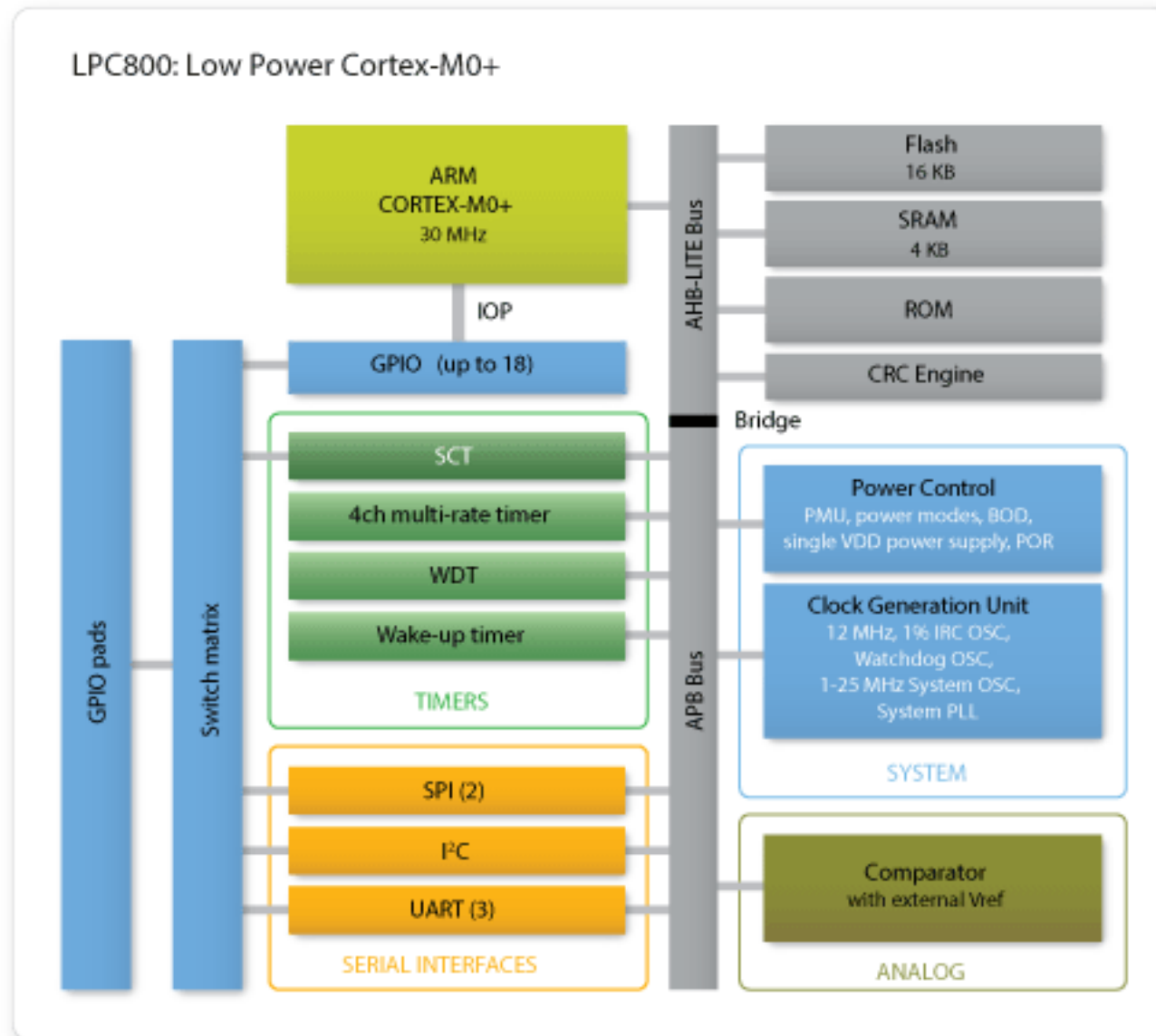
VSEL



# ARM Cortex M0+



# NXP LPC800 CM0+



# Cortex M0

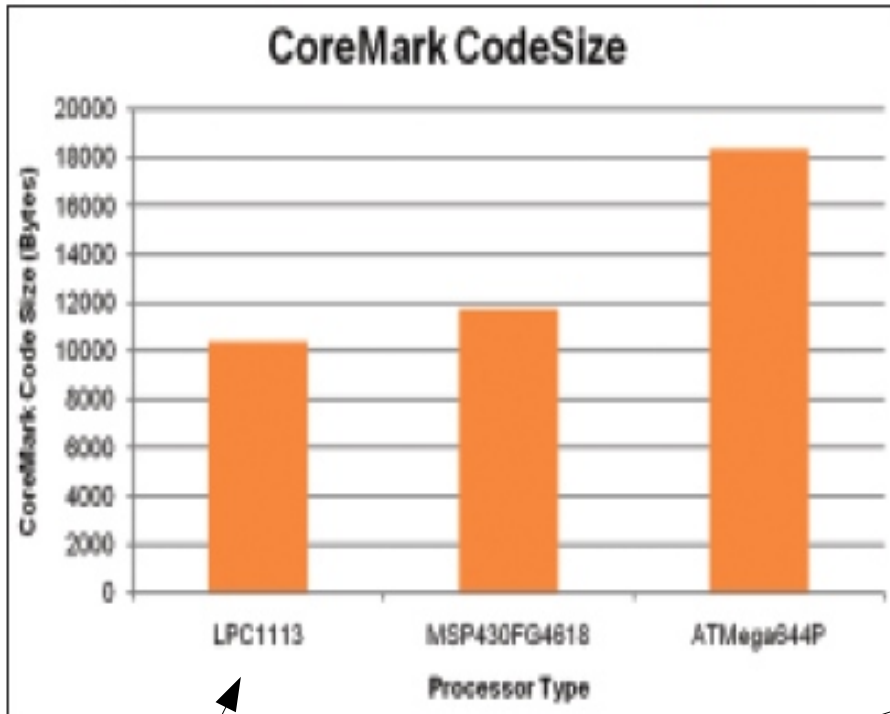


Figure 4: CoreMark code size optimized for fastest performance

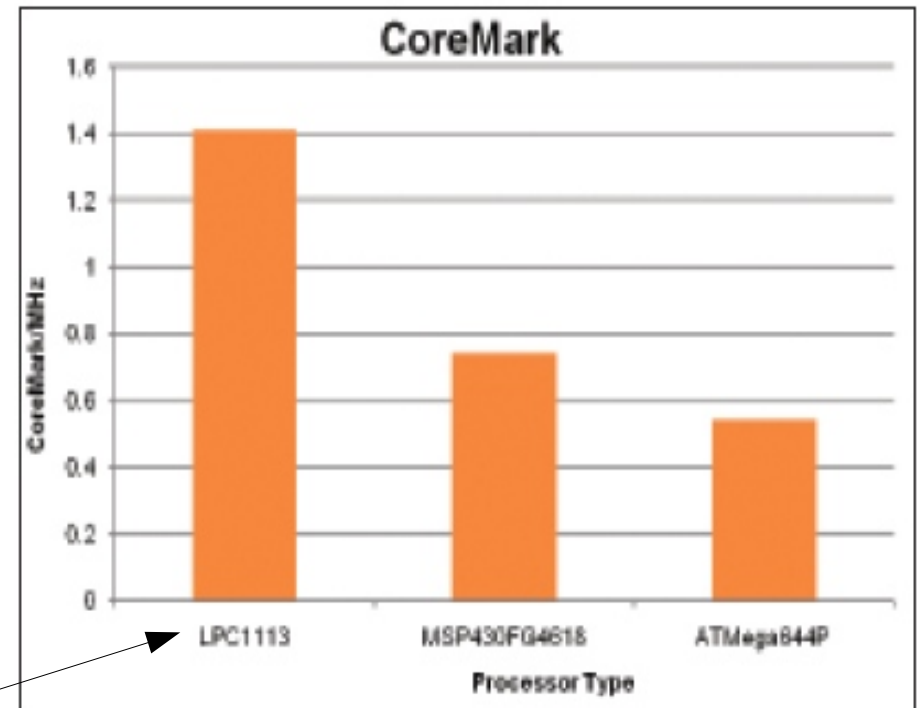


Figure 5: CoreMark Performance

Cortex M0

# Cortex M0(+) Instruction Set

## Just 56 Instructions !



## ARM Cortex-M0+ Features

|                        |                                                                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISA Support            | Thumb® / Thumb-2 subset                                                                                                                                                 |
| Pipeline               | 2 stage                                                                                                                                                                 |
| Performance Efficiency | 1.77 CoreMark/MHz - 0.93 DMIPS/MHz (ARM Compiler version 5.0.90)                                                                                                        |
| Memory Protection      | Optional 8 region MPU with sub regions and background region                                                                                                            |
| Interrupts             | Non-maskable Interrupt (NMI) + 1 to 32 physical interrupts                                                                                                              |
| Sleep Modes            | Integrated WFI and WFE Instructions and Sleep On Exit capability<br>Sleep & Deep Sleep Signals<br>Optional Retention Mode with <a href="#">ARM Power Management Kit</a> |
| Bit Manipulation       | Bit banding region can be implemented with <a href="#">Cortex-M System Design Kit</a>                                                                                   |
| Enhanced Instructions  | Hardware single-cycle (32x32) multiply option                                                                                                                           |
| Debug                  | Optional JTAG & Serial-Wire Debug Ports Up to 4 Breakpoints and 2 Watchpoints                                                                                           |
| Trace                  | Optional Micro Trace Buffer                                                                                                                                             |

## ARM Cortex-M0+ Implementation Data\*

|                   | 180ULL(7-track, typical 1.8v, 25C) | 90LP (7-track, typical 1.2v, 25C) | 40G(9-track, typical 0.9v, 25C) |
|-------------------|------------------------------------|-----------------------------------|---------------------------------|
| Dynamic Power     | 52 $\mu$ W/MHz                     | 11 $\mu$ W/MHz (9 $\mu$ A/MHz)    | 3 $\mu$ W/MHz                   |
| Floorplanned Area | 0.13 mm <sup>2</sup>               | 0.04 mm <sup>2</sup>              | >0.01 mm <sup>2</sup>           |



# Power Consumption

## Battery Lifetime

3  $\mu$ W / MHz @ 0.9V  $\rightarrow$  30 $\mu$ W @ 10 MHz  
(MPU Core only w/o memory)

Single Cell Lithium Battery Type AA: 5Ah \* 3.3 V = 16.5 Wh  
16.5Wh / 30 $\mu$ W = 550,000.0h = 62a

Sunlight: 1kW /m<sup>2</sup>

Room-Light: 100W/m<sup>2</sup>

Solar-Cell Output @ 10% Efficiency P/A = 10W / m<sup>2</sup>

Cell size needed for 30 $\mu$ W      A      = 30 $\mu$ W / 10W / m<sup>2</sup> = 3 \*10<sup>-6</sup> m<sup>2</sup>  
= 1.7 mm \* 1.7 mm

- $\rightarrow$  No Problem to run an application continuously over the complete lifetime from a single cell battery
- $\rightarrow$  Use of energy harvesting to feed the processor is simple and cheap
- $\rightarrow$  Creates the possibility to develop thousands of state-of-the-art IOT applications

# Cortex M0 Processor Features

- Single bus interface = Von Neumann architecture
- Tight integration of system peripherals  
reduces area and development cost
- Thumb instruction set  
combines high code density with 32-bit performance
- Support for single-cycle I/O access
- Power control optimization of system components
- Integrated sleep modes for low power consumption
- Fast code execution  
permits slower processor clock or increases sleep mode time
- Optimized code fetching  
for reduced flash and ROM power consumption
- Hardware multiplier (optional)
- Deterministic, ultra fast interrupt handling

# Configurability

|                                  |                              |
|----------------------------------|------------------------------|
| Interrupts External              | Interrupts 0-32              |
| Data endianness                  | Little-endian or big-endian  |
| SysTick timer                    | Present or absent            |
| Number of watchpoint comparators | 0, 1, 2                      |
| Number of breakpoint comparators | 0, 1, 2, 3, 4                |
| Halting debug support            | Present or absent            |
| Multiplier                       | Fast or small                |
| Single-cycle I/O port            | Present or absent            |
| Wake-up interrupt controller     | Supported or not supported   |
| Vector Table Offset Register     | Present or absent            |
| User/Privilege support           | Present or absent            |
| Memory Protection Unit           | Not present or 8-region      |
| Reset all registers              | Present or absent            |
| Instruction fetch width          | 16-bit only or mostly 32-bit |

# ARM MPU Registers

|           |                                                                                                                                                                                                                                                                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R0-R12    | R0-R12 are general-purpose registers for data operations.                                                                                                                                                                                                                                                                                     |
| MSP (R13) | The <i>Stack Pointer</i> (SP) is register R13. In Thread mode, the CONTROL register indicates the stack pointer to use, <i>Main Stack Pointer</i> (MSP) or <i>Process Stack Pointer</i> (PSP).                                                                                                                                                |
| PSP (R13) |                                                                                                                                                                                                                                                                                                                                               |
| LR (R14)  | The <i>Link Register</i> (LR) is register R14. It stores the return information for subroutines, function calls, and exceptions.                                                                                                                                                                                                              |
| PC (R15)  | The <i>Program Counter</i> (PC) is register R15. It contains the current program address.                                                                                                                                                                                                                                                     |
| PSR       | <p>The <i>Program Status Register</i> (PSR) combines:</p> <ul style="list-style-type: none"><li>• <i>Application Program Status Register</i> (APSR)</li><li>• <i>Interrupt Program Status Register</i> (IPSR)</li><li>• <i>Execution Program Status Register</i> (EPSR).</li></ul> <p>These registers provide different views of the PSR.</p> |
| PRIMASK   | The PRIMASK register prevents activation of all exceptions with configurable priority. For information about the exception model the processor supports, see <a href="#">Exceptions on page 3-10</a> .                                                                                                                                        |
| CONTROL   | The CONTROL register controls the stack used, and optionally the code privilege level, when the processor is in Thread mode.                                                                                                                                                                                                                  |

# Cortex M0 / M0+

- “Lightweight” version of the Cortex M Processor Family
- Subset of the instruction set of Cortex M3 / M4 / M7
- Ultra-low power versions available
- Reasonable code-efficiency and code-density
- User-configurable feature set